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Multifunctional Double-Network Self-Healable Hydrogel and Its Application to Highly Reliable Strain Sensors

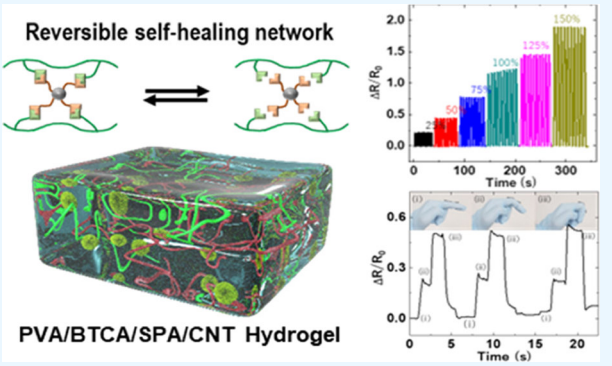
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ABSTRACT: Self-healable hydrogels present an emerging capability in energy harvesting, drug-release agents, artificial skin, and tissue engineering. Despite the various advantages of hydrogels, their low thermal stability, dehydration resistance, and mechanical properties hinder their practical applications. Herein, we introduced glycerol, 1,2,3,4-butanetetracarboxylic acid (BTCA), and sodium polyacrylate (SPA) into a hydrogel composed of poly(vinyl alcohol) (PVA)/agarose/borax. This resulted in the fabrication of a dual network hydrogel (DNH) that integrates attractive properties such as self-healing properties induced without physicochemical stimuli, stretchability, dehydration resistance, anti-drying capability, or anti-freezing capability. The DNH developed in this study maintains flexibility after storage for 1 h at −20 °C and 77.3% of its original weight after storage at 50 °C for 168 h, indicating its superior anti-freezing capability and water retentivity along with excellent self-healing properties. In addition, by blending carbon nanotubes (CNTs) to impart electrical conductivity, we have demonstrated that the CNT-embedded DNH can be successfully applied as an adhesive conductive medium for a stimulus-sensitive sensing channel of strain sensors, one of the key prospects of wearable electronic skins. In particular, the CNT-embedded DNH-based strain sensor can monitor human motion efficiently when attached to human skin without any skin trouble, even after being attached to the skin for several days. Our study can open an avenue for exploring core conductive adhesive hydrogel materials for next-generation wearable electronic devices.



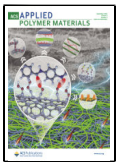
KEYWORDS: *self-healable hydrogel, multifunctionality, double-network, dehydration resistance, wearable electronics, strain sensor*

INTRODUCTION

The development of multifunctional, deformable, and biocompatible soft electronic material is one of the important key points to meet the growing demand for next-generation wearable electronics. Hydrogels, soft and moist materials that can swell by absorbing large amounts of water while maintaining their three-dimensional (3D) networks, have attracted attention as candidates for realizing ideal soft electronics. In general, hydrogels exhibit high biocompatibility with blood and human tissues without affecting the metabolic processes of the organism owing to their innate structure and composition, similar to those of the extracellular matrix.[1−3] These features enable the comprehensive implementation of hydrogels in various applications in biological medicine and tissue engineering fields, such as drug-releasing agents, enzyme encapsulation, protein electrophoresis, cell migration, artificial skin, and tissue engineering scaffold materials.[4−12] Moreover, hydrogels have unique intrinsic deformability, self-healing, and transparency characteristics, which are expected to be utilized in wearable elastic and ionotropic applications.[13−16] This has led to recent considerable efforts focused on the development of flexible and multifunctional electronics by compounding carbon-based materials (carbon nanotubes (CNTs) or graphene) to impart electrical properties to hydrogel materials.[17−24]

Despite the aforementioned advantages of hydrogels, the fabricated materials exhibit low thermal stability and limited mechanical properties for deformable and elastic applications due to structural approaches involving composites with inorganic materials.[21,25] To overcome these limitations, it is crucial to develop novel hydrogel designs offering excellent physical properties while maintaining biocompatibility and eco-friendliness. As a strategy to make this possible, the use of double-network hydrogels (DNHs) comprising polyvinyl alcohol (PVA), agarose, and sodium-tetraborate-decahydrate (borax), exhibiting significantly enhanced self-recovery, mechanical robustness, and strength, has been suggested to

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approach this problem.[26] In a DN-hydrogel (DNH) system, the hydrogen-bonding interactions in agarose gel robustly support the formation of the hydrogel and physiochemically cross-linked networks while exhibiting self-healing abilities induced by the hydrogen bonding of PVA and borate. The cross-links are readily associated and dissociated at room temperature, whereas the cross-linked network is affected by mechanical deformation.[27] Hydrogen bonds can break and reform spontaneously, allowing the hydrogel to self-heal and recover. Moreover, the PVA−borax hydrogel exhibits limited dimensional stability flowing under significant mechanical stress due to its non-Newtonian behavior. Therefore, the polymer chain and free borate ions enable sufficient mobility owing to the hydrogen bonding across the broken interfaces to initiate rapid self-healing without external stimuli. Notably, agarose, borax, and glycerol were added to enhance the stretchability and plasticity by forming a cross-linked network via massive hydrogen bonding.

These characteristics have recently triggered numerous investigations to develop DNHs, through an eco-friendly and low-cost synthetic process for use in various electronics-related applications, including conducting and resisting materials, electrodes, or sensors with particular functionalities and stimulus-responsive properties.[28−37] Tsai et al. fabricated a double-network hydrogel using PVA, agarose, borax, zinc oxide (ZnO), and poly(3,4-ethylenedioxythiophene):poly-(styrenesulfonate) (PEDOT:PSS).[29] Using the latter as a sensing layer, they fabricated a self-powered, self-healing, and wearable ultraviolet (UV) photodetector. With the DN-hydrogel substrate integration, the photodetector restored up to 90% of its initial efficiency even after 5 healing cycles and showed a very rapid self-healing within 10 s.

As previously described, DNHs have recently become the focus of extensive research and development. However, the thermal drying properties of traditional DNHs significantly narrow the temperature range of their application and lower their lifetime stability.[38] For example, at elevated temperatures, hydrogels dry quickly due to dehydration, whereas at sub-zero temperatures, their moisture content freezes, which alters their physical properties and limits their use.[39] Furthermore, hydrogel dehydration occurs with time or by exposition to dry conditions. Thus, the physical stability of hydrogels depends on the service time and temperature of the environment to which they are applied.[38,40] It follows that maintaining the stability of shape and physical properties depending on the temperature is the key to securing the extended applicability of DNHs in various ubiquitous environments. Thus, an adequate strategy must be developed to realize hydrogels with enhanced dehydration resistance while retaining thermal sensitivity. Previous studies attempted to increase dehydration resistance by adding an antifreeze, such as glycerol or ethanol, to the hydrogel.[41,42] Liu et al. prepared a hydrogel with anti-drying capability and anti-freezing capability using a simple physical mixing of glycerol and a one-time freeze−thaw process in the fabricating of the PVA/gelatin hydrogel.[41] The prepared hydrogel showed a higher anti-drying characteristic than conventional hydrogels under a temperature range from room temperature to 80 °C due to glycerol. The flexibility of the hydrogel was maintained even after storage in an environment of −25 °C. Nonetheless, the anti-drying capability of this hydrogel with added glycerol or ethylene glycol remained insufficient.

In this study, we added glycerol, 1,2,3,4-butanetetracarboxylic acid (BTCA), and sodium polyacrylate (SPA) to a hydrogel composed of agarose, PVA, and borax, thereby developing a new class of stretchable and dehydration-resistant self-healing DNH, composed mainly of the biodegradable agarose. We demonstrate its application in strain sensors by embedding CNTs to the developed DNH as the conductive sensing medium. Our developed conductive DNH exhibited rapid self-healing, restoring 98% of its initial conductivity level within 5.5 s, which does not necessitate any external stimuli such as heat, pH, light, or catalysts. In addition, the rapid cohesive viscoelasticity of this self-healable CNT/DNH possesses the superior self-recovery character of the electrical properties during relation and contraction processes under mechanical stress, which is efficient at monitoring external stimuli such as strain, crease, and distortion, when applied as a strain sensor. Exploiting these critical advantages, we showed that movement could be accurately detected by attaching the developed sensor to the skin. The CNT/DNH developed in this study could be utilized as a practical, resilient, and versatile surface-attachable sensing medium for next-generation biodegradable electronics such as e-skin applications.

EXPERIMENTAL SECTION

Synthesis of DNH. PVA (1.5 g, M_w 13,000−23,000, Samchun Chemical) and agarose (100 mg, Type I, Sigma-Aldrich) were dissolved in 7.65 mL of water and 850 mg of glycerol (99%, Samchun Chemical) at 90 °C for 2 h. The solution was cooled to room temperature when the PVA was completely dissolved. Then, 20 mg of BTCA (99%, Sigma-Aldrich) and 50 mg of SPA (average M_w ∼5100, Sigma-Aldrich) were added to the prepared solution at 25 °C. As BTCA and SPA were utterly dissolved, 0.8 wt % of CNT (multiwalled, 40−60 nm, 5−15 μm, TCI Chemical Co.) was added to the prepared solution, and sonication was performed for 20 min for dispersion, then 500 mg of borax (Samhyun Pharm, Korea) was added.

Characterizations of Hydrogels. The hydrogels were stored at a constant temperature of 25 and 40 °C for 7 days to analyze and compare their anti-drying properties. The value of $W_t/W_{t=0}$ was used to indicate the degree of resistance to dryness for the hydrogels at different storage times. The $W_{t=0}$ value is the initial mass of the hydrogels, and the W_t value is the mass of hydrogels for different storage times. Furthermore, the anti-freezing properties of the hydrogels were observed and compared in terms of freezing behavior after being stored at −20 °C for 1 h. The cooled and/or frozen hydrogels were subjected to deformation by hand, including bending to ascertain the state. The self-healing process of the hydrogel was observed using an optical microscope (OM, Hirox KH-7700). The hydrogel film was sliced using a razor blade, and then the crack of the sample was continuously observed under the same microscope. The biodegradability of the hydrogels was evaluated after the samples were stored in a container at room temperature for 15 days under 70% humidity. Mechanical strength and stretchability were assessed using a force gauge (Mark-10, Electromatic Equipment Co., Inc.) The hydrogel sample was prepared (3 cm × 1 cm × 0.3 cm), and tensile strength and elongation at break were recorded according to the applied force.

Electrical Characterization. The conductivity of the hydrogels was assessed using an Agilent HP4156A Precision Semiconductor Parameter Analyzer at room temperature under ambient conditions. The 2.5 cm × 0.5 cm × 0.3 cm hydrogel samples were sandwiched between two aluminum electrodes in a two-electrode system. The following equation can define the conductivity(σ)

$$\sigma = \frac{1}{R}\frac{t}{A}$$

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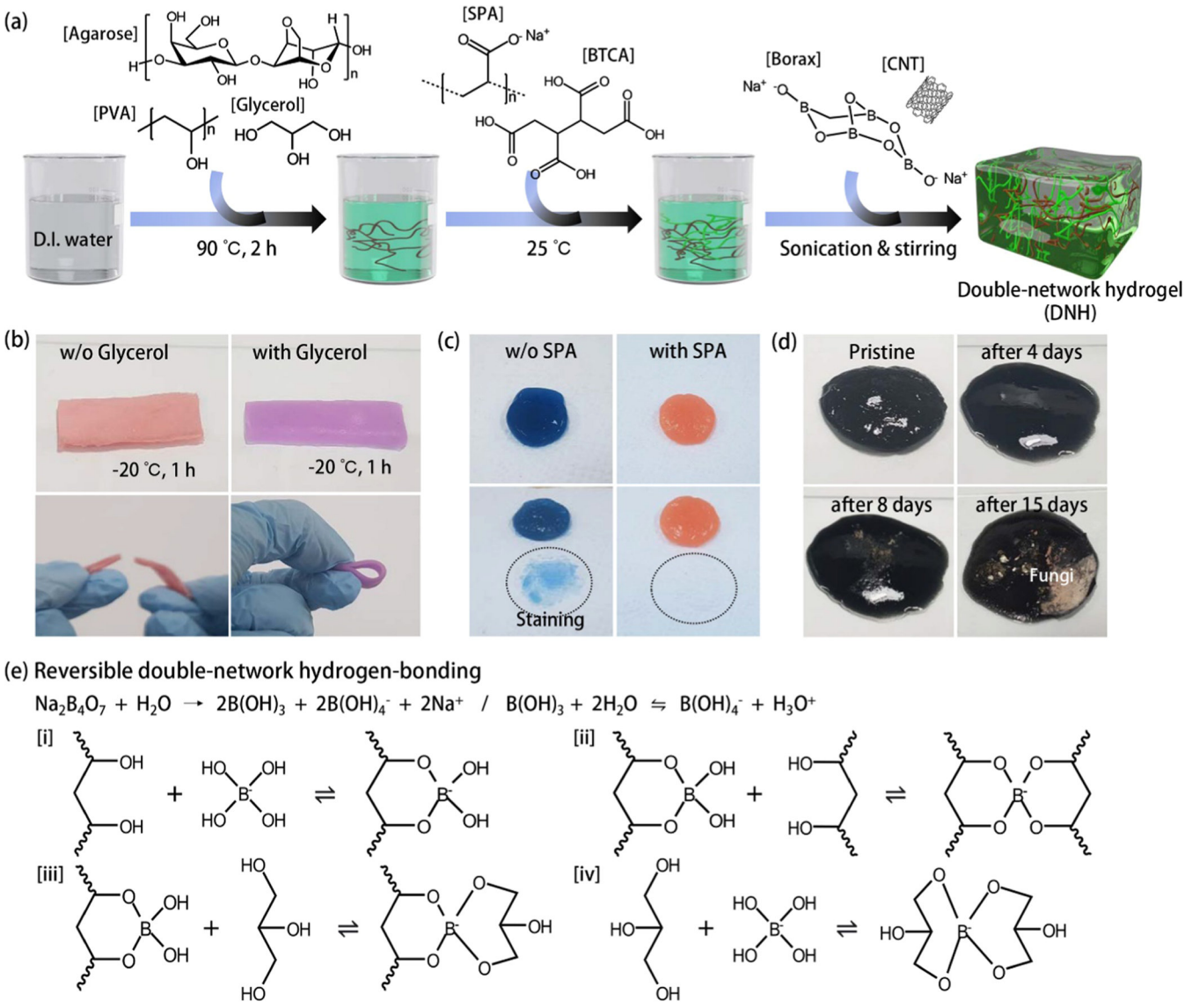


Figure 1. Preparation and characterizations of the CNT-embedded DNHs. (a) Schematic illustration of the preparation of the CNT-embedded DNHs. (b) Flexibility evaluation of DNH according to the addition of glycerol after being kept at −20 °C. (c) Moisture-retention properties of DNH according to the addition of SPA. (d) Photographs for verification of the biodegradable properties of the developed DNHs. (e) Reversible cross-linking process induced by the DN hydrogen-bonding mechanism.

where *t*, *A*, and *R* were the hydrogel thickness in cm, the electrode area in cm^2, and the hydrogel resistance in Ω, respectively. The response of the strain sensor was assessed by measuring the resistance using an Agilent 4284A Precision inductance, capacitance, and resistance (LCR) meter at room temperature under ambient conditions. The applied voltage to obtain the *R* values was fixed as a voltage of 1 V and a sweeping frequency of 1 MHz. The strain-sensing capabilities of the sensor were characterized using a custom-designed sensor probe station outfitted with a programmable *x*- and *y*-axis moving stage controlled by a stepper motor (DRV-014, Thorlabs, Inc.) with 50 nm resolution, all interfaced through a computer. The hydrogel formed a sample (3 cm × 1 cm × 0.3 cm) that connected with two wires to make a simple resistive strain sensor.

Modeling the Hydrogel Microstructure. The hydrogel microstructure was modeled using commercial software, Materials Studio (Dassault Systèmes), and open-source code, LAMMPS (Sandia Lab.). Initially, PVA, glycerol, and CNT were randomly located in a periodic cell with a density of 0.35 g/cc. Molecular components have a weight fraction of 0.56:0.32:0.12, respectively. PVA chain comprises 224 monomers and has a molecular weight of 9870 g/mol. An armchair-type CNT (5, 5) with a length of 6 nm is considered. The modeled unit cell was equilibrated at 300 K for 200 ps using an NVT ensemble, followed by geometry optimization by the conjugate gradient method. Water molecules were then inserted into the free space. Insertion sites of water molecules were randomly selected every 0.1 ps, and water molecules were newly created if no other atoms were present within a radius of 2 Å around the selected position. After the insertion was completed, water molecules accounted for 60.8 wt % of the system. Finally, the hydrogel was equilibrated with the NPT ensemble at 300 K and 0.1 MPa for 300 ps. All interatomic interactions were described using a polymer consistent force field (PCFF) with a timestep of 0.5 fs [R1]. The hydrogel microstructure was stretched by 150% with an engineering strain rate of 10^{10}/s under the NPT ensemble at 300 K and 1 atm. After deformation, a surface mesh with a probe radius of 5 Å was applied to all organic components to visualize the internal microstructure.

RESULTS AND DISCUSSION

Figure 1a illustrates the fabrication process of conductive CNT-embedded DNHs, mainly comprising PVA and agarose for imparting biodegradability and eco-friendliness. The PVA

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and agarose are the matrix polymers constituting a hydrogel and form a hydrogel through a cross-linking reaction with borax. The hydrogels prepared using only PVA and agarose exhibit a deficient viscoelastic character, inferring breakage upon stretching. To prevent this inconvenient behavior, glycerol is added to strengthen the viscoelasticity and ductility, further securing the anti-freezing properties by lowering the freezing point. To verify these characteristics, we compared the behaviors of the hydrogels with and without added glycerol, as shown in Figure 1b. After 1 h at −20 °C, the hydrogel without glycerol exhibited brittle rigidity, indicating a frozen state.

In contrast, the glycerol-added hydrogel retained its flexibility under similar environmental conditions, demonstrating superior flexibility. Further, 1,2,3,4-butanetetracarboxylic acid (BTCA) and sodium polyacrylate (SPA) were added to improve the drying resistance by increasing hydrogen-bonding production with water molecules through many carboxyl or hydroxyl groups in their molecules, resulting in the reduction of water molecules escaping to the outside. Lavric et al. reported that BTCA is a cross-linking agent for synthesizing hydrogel composites with holdable humidity by screening the temperature and enhancing the wearing comfort.[43,44] Moreover, the pH and thermo-sensitivity can be adjusted to enable the controlled delivery of drugs and moisture, facilitating the application of these composites for wound treatments. The SPA forms the ionic interaction between the surrounding water molecules and carboxyl or hydroxyl groups in bent molecules.[45] Thus, it stably holds water molecules inside the hydrogel and prevents water from leaking. As shown in Figure 1c, the moisture-retaining ability of the developed hydrogel was tested in the presence and absence of SPA. In the latter case, the image (left, blue) shows a distinct strain induced by the leaking water, while no mark was observed in the SPA-containing hydrogels (right, orange). This indicates enhanced dehydration resistance after SPA addition. Moreover, CNTs were added to the DNHs to simultaneously achieve excellent electrical conductivity and a skin-attachable sensory device.

Borax was used to induce hydrogelation through hydrogen bonding with PVA, agarose, and glycerol, as described above, which induced self-healing ability. A boric-acid/tetra-functional borate ion forms when borax is immersed in water. A tetra-functional borate ion formed during this process induces several hydrogen bonds with the surrounding hydroxyl groups, producing a 3D-like-cross-linked network. In contrast to covalent bonds or esterifications, these hydrogen bonding are suitable for securing the self-healing properties of the hydrogels by reversible recombinations.[43−45] In particular, we structured DNH by adding glycerol and BTCA to generate massive hydrogen bonding, thereby maximizing the viscoelastomeric character (stretchability and resilience) based on the superior self-healing ability.

The biodegradability of CNT/DNH was experimentally evaluated, as shown in Figure 1d. The results confirm the spread of the mold after keeping the hydrogel for 15 days in 70% humidity. Additionally, despite the high hydrogen-bonding density of the developed DNH, biodegradability was obtained owing to the eco-friendly materials used. The reversible cross-linking processes induced by the DN hydrogen-bonding mechanism proposed in this study are shown in Figure 1e.

Although we confirmed the anti-freezing capability, water-holding ability, and biodegradability of DNH shown in Figure 1, it is necessary to clarify the degree of its long-term water-retention at operating temperature or under harsh conditions. This points towards the importance of thermal stability at the operating temperatures in determining practical applications.[46,47] We monitored the thermal resistivity and moisture-retention properties of DNHs through time-dependent weight variations during long-term storage. Four types of DNHs were prepared for this evaluation (Table 1). Sample 1 is a general hydrogel prepared by adding only PVA, agarose, and borax as a hydrogen-bond-based crosslinker. Sample 2 is a hydrogel with flexibility and anti-freezing capability, obtained by adding glycerol to sample 1. Sample 3 is a case where BTCA is added to sample 2 to form the DN of the hydrogel. Finally, sample 4 improved the water retention of the DNH by adding SPA to sample 3. Figure 2a shows the weight changes of the four prepared samples for 168 h (7 days) to evaluate the thermal stability at room temperature (i.e., 25 °C). As seen in the figure, the simple hydrogel (sample 1) composed of PVA and agarose maintained 88.4% weight after 168 h, and it was confirmed that 91.1% of the weight was maintained in sample 2 with glycerol added.

Table 1. Content Information of Hydrogels Prepared in this Study

	contents of hydrogel
sample 1	PVA + agarose + borax
sample 2	PVA + agarose + glycerol + borax
sample 3	PVA + agarose + glycerol + BTCA + borax
sample 4	PVA + agarose + glycerol + BTCA + SPA + borax

On the other hand, under the same experimental conditions, Samples 3 and 4 maintained 94.2 and 95.2% of their weights, respectively. These results prove that the water retention inside the hydrogel is greatly improved by the formation of the DNH structure. The enhancement of the water retention of DNH could be further emphasized when the weight change is confirmed under harsh conditions of 50 °C, as shown in Figure 2b. Results show that samples 1 and 2 stored at 50 °C for 168 h maintained only 33.7 and 55.6% of their weights, respectively. Although the water retention of the simple hydrogel structures has been improved by adding glycerol, the aforementioned levels remain unsuitable for practical applications in wearable or adhesive electronics. However, much better water-retention characteristics were observed for samples 3 and 4. Sample 3, to which BTCA was added to form DN inside the hydrogel, maintained 70.7% of its original weight even after keeping at 50 °C for 168 h, whereas sample 4, with added SPA forming the ionic interaction inside a DNH, maintained 77.3% of its weight. Although the DNH introduced in this study does not show perfect water retention in a high-temperature environment, it maintained 77.3% of the original weight for a week under the harsh conditions of 50 °C, which is worth taking advantage of for a wide range of general applications. Figure 2c compares the stretchability or deformability after keeping the samples at 50 °C for 48 h to emphasize the difference in usefulness owing to the water retention of the four samples.

As described above, sample 4 presented the best water-retention characteristics at high temperatures owing to the DNHized structure and the ionic interaction inside the DNH formed by SPA molecules. Based on these results, the self-healing properties of this sample were thoroughly analyzed (hereafter, sample 4 is referred to as DNH). Figure 2d shows

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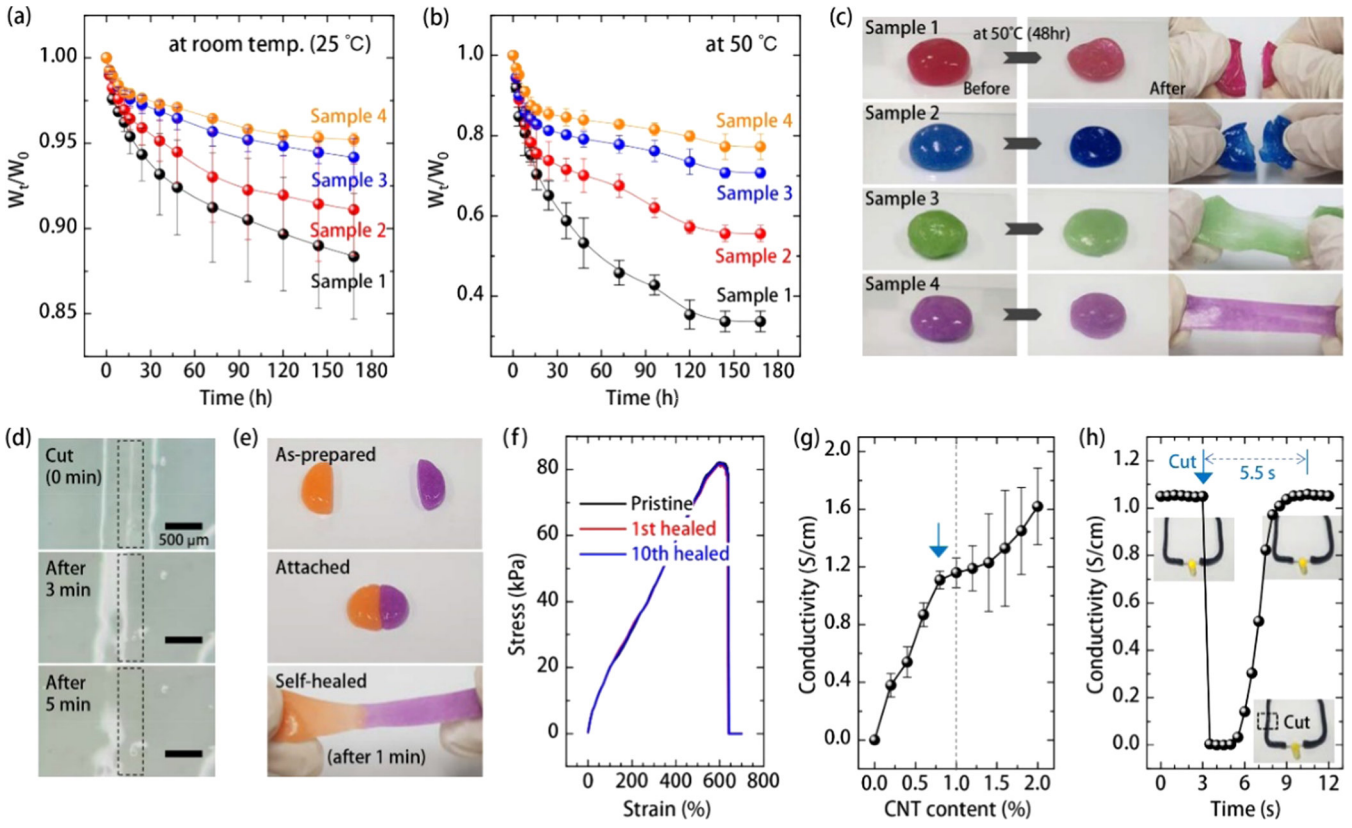


Figure 2. Long-term water retentivity of DNHs. Time-dependent weight changes of the samples under various conditions at (a) 25 °C and (b) 50 °C. (c) Stretchability test of samples prepared under different conditions after standing at 50 °C for 48 h. (d) Demonstration of self-healing ability for the prepared DNHs. Scale bar is 500 μm. (e) Optical microscopy to observe the self-healing process of the separated hydrogel. (f) Stress−strain curves of repetitive self-healed DNHs. (g) Electrical conductivity change curve with varying content of CNT contained in DNHs. (h) Electrical recovery test of the CNT-embedded DNHs and demonstration of LED electrode connection.

the representative optical microscopy images of the DNH healing process after being severed by a blade. The cut part of the DNH gradually returned to its original state after cutting and then fully recovered after 5 min at room temperature without any external stimulation. Moreover, when two distinct DNH pieces were attached and stretched for 1 min, they behaved like one piece, which is another result demonstrating the self-healing ability shown in Figure 2e. Then, the stress−strain characteristics were compared to check how the recombination strength after self-healing differs from the original state. Figure 2f shows the mechanical properties of the DNH pristine state after the first self-healing, which was repeated 10 times. As shown in Figure 2f, the developed DN-hydrogel shows consistently similar stress−strain curves even for all repeated tests, which proves the superiority of its self-healing properties.

Our developed self-healable, anti-drying, and anti-freezing DNH exhibits a high potential for applications as a conductive and adhesive medium. A conductive DNH was prepared for this application by blending and dispersing CNTs. First, the conductivity according to the CNT content was analyzed to determine the optimal CNT content. Figure 2g shows that the DNH conductivity increases with the CNT content (0−2 wt %). Interestingly, for a CNT content beyond 1 wt %, the deviation of conductivity was confirmed to increase significantly, which is associated with the nonuniform dispersion of CNT in the hydrogel. Despite excellent conductivity in the cases of the CNT content of 1 wt % or more, we estimated that this conductivity would not be reliable due to the nonuniform dispersion. Thus, we selected a case with a CNT blending ratio of 0.8 wt %, as indicated by the arrow. The ability of our developed DNH to recover its conductivity through self-healing after cutting was tested and evaluated, as shown in Figure 2h. Initially, the conductivity of the CNT-embedded DNH was approximately 1.05 S/cm. The results show that the conductivity fully recovers rapidly within 5.5 s after cutting. This indicates that, although the CNT was contained in the DNH, its conductivity was quickly restored by inducing a self-healing process. The result could be visually confirmed by connecting a light-emitting diode (LED) device, as shown in Figure 2h.

In addition to many noticeable capabilities, such as self-healing ability, stretchability, anti-drying and anti-freezing capability, and water-retention of the CNT-embedded DNHs, we further examined the feasibility of strain sensors to detect human motion. The resistance values of CNT-embedded DNHs increase with tensile strain owing to the enhanced path length in the cross-linked polymer network and between the CNTs during tensile testing.[48,49] The following equation can define the strain sensitivity (GF)

$$GF = (R - R_0)/\varepsilon R_0$$

Dynamic performance behavior, drift, and overshoot measurements of the CNT-embedded DNH strain sensor subjected to step-wise increases or decreases in tensile strain were investigated, as shown in Figure 3b. Here, ε is the strain and R_0 is the resistance at 0% tensile strain. As shown in Figure 3a, when the tensile strain degree was increased without

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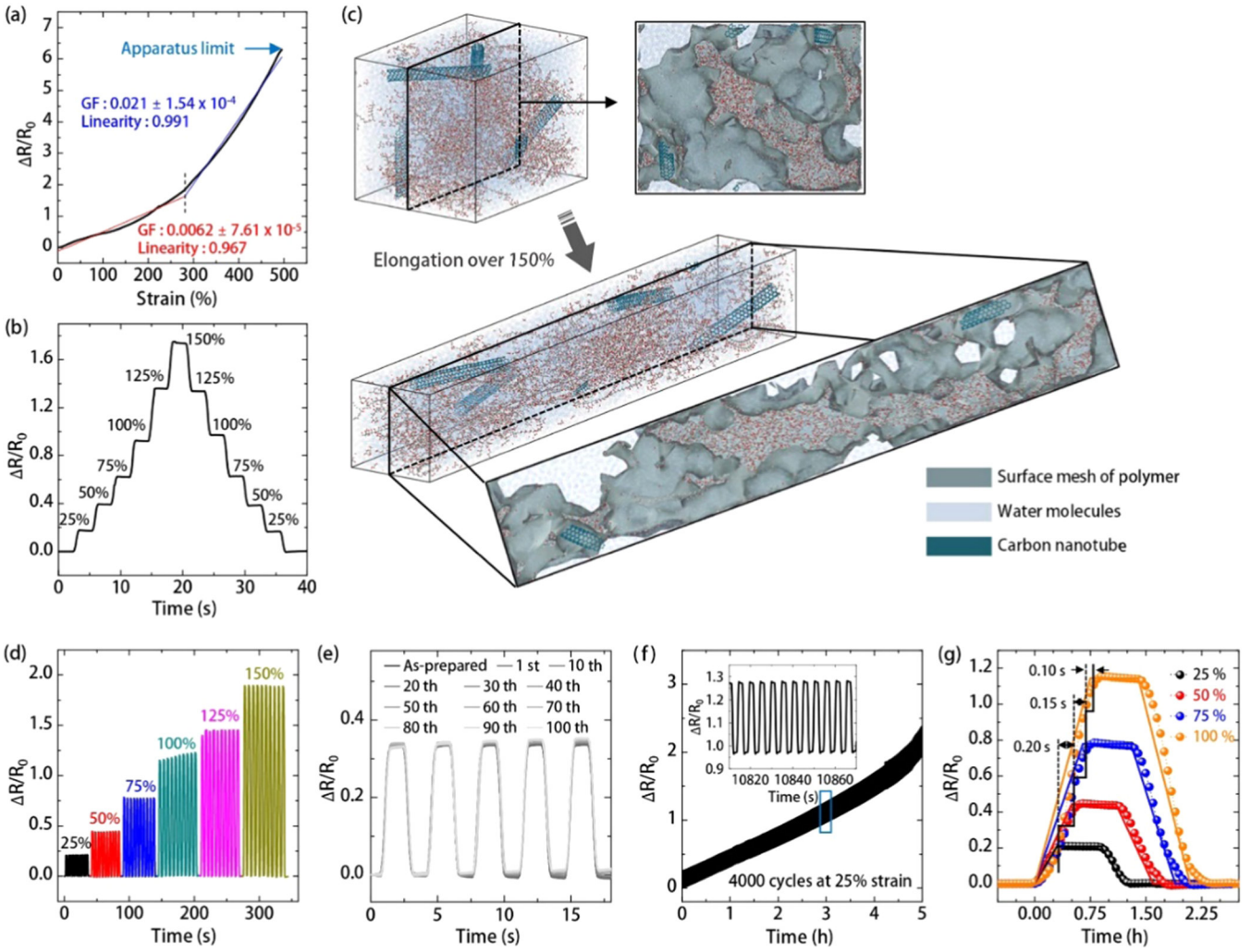


Figure 3. Strain sensor based on CNT-embedded DNHs. (a) $\Delta R/R_0$ as a function of strain values up to 500%. (b) Step-by-step repeatability test in which $\Delta R/R_0$ was measured over repeated stretching/releasing cycles from 0 to 150%. (c) Simulated image of the CNT-embedded DNH at the applied strain. (d) $\Delta R/R_0$ of the sensor as a function of time during which the strain applied was increased step-wise from 25 to 150% with a 40 s holding time at each strain level. (e) $\Delta R/R_0$ variations were applied along various tested repetitive healing processes. (f) Stability test of the $\Delta R/R_0$ response for 4000 cycles at 25% strain. (g) Response times of the samples under different strains from 25 to 100%.

considering the recovery character, the relative resistance increased with strain. The variation in the resistance depending on the tensile strain shows that at a low strain degree the GF and linearity were 0.0062 ± 7.61 × $10^{-5}\%^{-1}$ and 0.967, respectively, whereas at a high strain degree, the GF and linearity were 0.021 ± 1.54 × $10^{-4}\%^{-1}$ and 0.991, respectively, based on 280% of the tensile strain. The variations in the resistance of CNT-embedded DNHs as a function of the tensile strain show results measured only in the range of 494% due to equipment limitations. Our sensor showed excellent operational reliability and stability in the measurement of the change in $\Delta R/R_0$ throughout the tensile strain range between 0 and 150%, with a holding time of 2 s at each strain level.

Figure 3b demonstrates measurement results of 150% of strain degree. The limitation of applying CNT-embedded DNH as a stable strain sensor in stretching and releasing was found to be up to approximately 160%. Recovery became nonelastic for a strain degree beyond this limit, which renders it unsuitable for application as a sensor. Here, a question arose during the experiment on the dynamic performance behavior of the strain sensor. Why would the strain sensor operate properly up to a strain level of about 160%, whereas the developed CNT-embedded DNH shows a somewhat constant increase in the resistance to the tensile strain up to almost 500%? To theoretically elucidate the answer to this question, we simulated the phenomenon that occurs when strain is applied to the CNT-embedded DNH using classical molecular dynamics simulations, as illustrated in Figure 3c. The mechanical properties of the calculated unit cells of the CNT-embedded DNH filled with excessive water molecules were considered. The unit cell was stretched up to 150% while preserving each molecular-scale component's chemical bond length, and the polymer chains' volume change due to mechanical deformation was visualized. Although water molecules occupy a specific volume inside the CNT-embedded DNH in an unstrained state, the surface mesh of the polymer is uniformly distributed throughout.

On the other hand, when a tensile strain of 150% or more is applied, the portion filled with the polymer is sharply narrowed. This induces a nonuniform distribution of hydrogen bonding between the polymers and water molecules constituting the DNH, which could be expected to weaken the deformation resilience.

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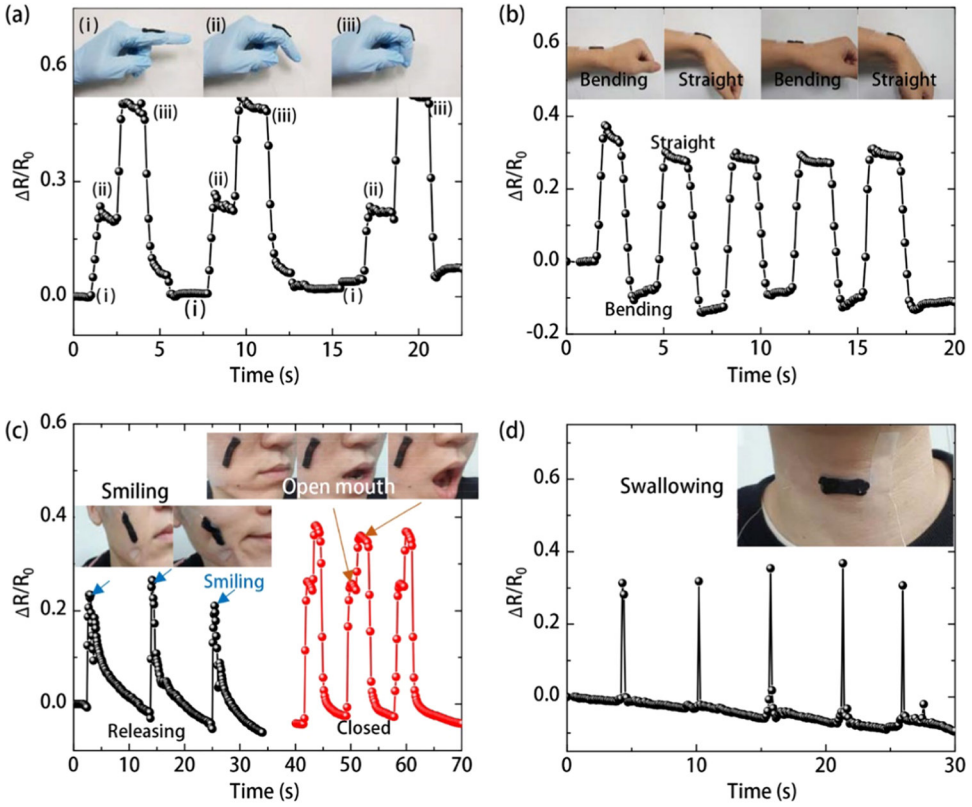


Figure 4. Demonstrations of the strain sensor attached to the human body. $\Delta R/R_0$ variations of the fabricated strain sensor with different motions attached to the (a) finger joint, (b) wrist, (c) cheek, and (d) neck. Various bending motions are performed on a body part that specific strain can be applied to the sensor, and the change of electrical resistance is observed.

Figure 3d shows the excellent response of $\Delta R/R_0$ values when our CNT-embedded DNH sensor was subjected to 10 cycles of stretching and releasing during strain increase from 25 to 150%, demonstrating the potential of applications to cohesive and wearable sensory systems. In the proposed 25−150% strain range, GF for the strain sensor presents $0.013 \pm 5.67 \times 10^{-4}\%^{-1}$, which indicates the improved response performance as a sensor compared to the value shown by simply stretching in Figure 3a. In addition, the linearity is 0.991, confirming that the sensing accuracy is excellent. To evaluate the reliability of the strain sensor to which the CNT-embedded DNH is applied, Figure 3e shows the $\Delta R/R_0$ results expressed by overlapping 10 repeated units among the 100 repeated measurements at 25% strain degree. Results demonstrate that there are no discernible losses.

Moreover, the $\Delta R/R_0$ response characteristics remained stable for 4000 cycles at 25% strain degree, as shown in Figure 3f. Although the baseline of the signal response continues to rise, this is a phenomenon often observed in flexible pressure or strain sensory studies, which is one of the issues that need to be improved in the future. Finally, the accuracy of the strain response signal was analyzed by comparing a single cycle at various strain degrees, i.e., 25, 50, 75, and 100%. As confirmed in Figure 3g, the response time of exactly 0.05 s appears for every 25% increase in the strain degree. In general, the sensing rate of the strain sensor has been quantified through the analysis of a single cycle, and it could be interpreted that the response and release times increase according to the strain degree if only this result is considered. However, considering the time required to deform the sensing medium (CNT-embedded DNH) in the sensor, which increases proportionally with the tensile strain, it could be determined that the response and release times are equally normalized.

Owing to the comprehensive advantages such as unique softness, elasticity (or stretchability), self-healing, and non-toxicity, the conductive CNT-embedded DNHs can be applied as a sensory system to construct biocompatible electronic skins. Figure 4a shows one of the applications as a strain sensor to detect bending by attaching a CNT-embedded DNH sensing medium to the knuckle of a finger. In the test, the index finger was repeatedly bent and straightened in a step, with the finger held in each bent state for several seconds. As the finger is bent, the $\Delta R/R_0$ increases owing to the stretching of the fixed sensor induced by bending; whereas when the finger is straightened, the $\Delta R/R_0$ decreases due to the release of the tensile strain. Our CNT-embedded DNH easily adhered to the surface of the finger skin because of its high flexibility and adhesiveness.

Similarly, the results of measuring the degree of bending by attaching it to the wrist are shown in Figure 4b. These results, consistent with those of the strain-sensing performances discussed above, indicate that our sensors are attractive candidates for applications in robotics and sensing devices to detect human joint movements. In addition, our strain sensor successfully detected smiling or mouth opening after attaching to the cheek and swallowing after attaching it to the uvula in Figure 4c,d, respectively. In particular, the CNT-embedded DNH introduced in this study did not cause any skin trouble even after being attached to the skin for several days.

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CONCLUSIONS

In summary, we demonstrated a new class of DNH integrating attractive properties such as self-healing ability induced without any physicochemical stimuli, stretchability, dehydration resistance, anti-freezing capability, and biodegradability. The DNH is mainly composed of PVA and agarose for imparting biodegradability and eco-friendliness. Mechanical properties (flexibility or stretchability) and dehydration resistance are strengthened by adding glycerol, BTCA, and SPA to enhance hydrogen-bonding production with water molecules. As a result, along with excellent self-healing properties, the flexibility was maintained even after storage for 1 h at −20 °C, and 77.3% of its original weight was maintained even after storage at 50 °C for 168 h, which indicates the superior anti-freezing and water-retention capabilities. To confirm the potential key prospects of our developed DNH as a conductive and adhesive medium for a stimulus-sensitive sensing channel, the electrical conductivity was demonstrated to recover within only 5.5 s after cutting fully. As a result of applying the CNT-embedded DNH as a sensing medium of the strain sensor, we confirmed the excellent sensing performances showing a GF and linearity of $0.013 \pm 5.67 \times 10^{-4}\%^{-1}$ and 0.991, respectively, up to a tensile strain level of 150%. In addition, the sensing performance maintained accuracy and reliability even after 100 repeated measurements and continuous operation of more than 4000 times. Finally, the CNT-embedded DNH-based strain sensor was capable of monitoring human motion efficiently when attached to the human skin, one of the wearable sensory system application areas, without any skin trouble even after several days of application to the skin. Consequently, these biodegradable CNT-embedded DNH with physical and electrical properties can be expected as a core adhesive material for the next-generation wearable electronic devices.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsapm.2c00902.

$\Delta R/R_0$ variations of the CNT-embedded DNH strain sensor with different bending angles attached to the finger joint (PDF)

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Notes

The authors declare no competing financial interest.

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